2025 International Conference on Information and Communication Technology for Development for Africa (ICT4DA) | 979-8-3315-9184-7/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICT4DA67218.2025.11282709

Potato Leaf Disease Detection using Transfer Learning Models And Grad-CAM++ Explainable XAi

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***Abstract*—This research study presented a method for detecting potato leaf diseases using transfer learning models and explainable XAI. It aimed to detect potato leaf diseases by comparing pre-trained transfer learning models of EfficientNet-B7, VGG16, VGG19, and ResNet50. Our transfer learning models are trained on a total sample of 3000 images of potato leaves with potato early blight (1000 images), potato late blight (1000 images), and healthy leaves (1000 images) with the same hyper-parameters. After extensive experimentation with 10% for testing, 10% for validation, and 80% for training and refinement of our models, the EfficientNet-B7 model outperformed the other models with an accuracy of 99.95% on the testing dataset. Finally, we integrated Explainable XAI of Grad-CAM++ for this model of EfficientNet-B7 to visualize, understand, and interpret which parts of potato leaf images contribute most to the model prediction. This has significant potential to enhance early disease identification, reduce crop losses, and increase food security in the world.**

***Index Terms*—Grad-CAM++,EfficientNet-B7,VGG16, VGG19,ResNet50,Transfer Learning,Potato Disease Detection**

I. Introduction

Agriculture is the backbone of economic development, and one of the agricultural crops is the potato, which provides an economic livelihood and nutrition for millions of people [1]. However, these crops are highly susceptible to early blight and late blight, which may be caused by viruses and bacteria that can significantly impact quality and yield [2] [3].

Nowadays one of the well-known types of disease detection for potato crops is eye observation, which has problems such as being prone to error, often subjective, time-consuming, and labor-intensive [4] [5]. One of the best mechanisms for potato disease detection is using computer vision, machine learning, and artificial intelligence, which has great potential for changing the trend of eye observation [6]. These methods can accurately and quickly detect these diseases without being labor-intensive or prone to error. A subfield of machine learning known as deep learning has shown exceptional abilities in pattern recognition and image analysis [7] [8].

Although existing deep learning models have an impressive ability in image analysis and pattern recognition, their understanding, interpretability, and explainability still need further investigation [8]. One of the mechanisms for this method is the explainable AI (AI) that has a great impact on understanding and visualization [8]. In order to help academics, practitioners, and end users understand why a specific classification or prediction was made, XAI aims to clarify the decision-making process of complicated models. By integrating XAI with the deep learning models, it creates human-understandable explanations for the imaging-based AI application in a variety of fields, such as computer vision, health informatics, and many other fields [8] [9].

Therefore, this research study proposed to detect potato leaf disease at an early stage using the transfer learning model of EfficientNet-B7, which is a member of the EfficientNet-B0 family [10], and Grad-CAM++, which is a member of the Grad-CAM family [11], explainable XAI by comparing it with models of VGG19 [12], VGG16 [12], and ResNet50 [13]. After we compared our models, EfficientNet-B7 demonstrated the highest accuracy, recall, precision, and F1 score among the models tested.

Finally, we integrated explainable XAI of Grad-CAM++ [11] with an EfficientNet-B7 [10] model to enhance the

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explainability and interpretability of the model. This approach not only improves the performance of the model but also increases the robustness, truth, and decision-making process for the farmers and agro-experts to highlight the most infected region of the potato image.

II. Related Works

The authors [14] stated that potato leaf disease detection using transfer learning of VGG16 and Grad-CAM explainable AI (XAI) [11] with a total sample of 2,152 images, which were sourced from the public availability of the village plant dataset. After extensive refinment of the models and experimentation , the researchers found an accuracy of 98% for testing and 97% for validation. For visualization,understanding and interprating , the model was integrated with Grad-Cam++.

This research article presented potato leaf disease detection with assessment of its severity [15] . The researchers collected a total of 4000 images of potatoes (early blight, healthy, and late blight) from the Oromia region, Jima Zone, Ethiopia, using smartphones. For this research study, the researchers compared AlexNet and VGG16 models for disease detection and severity. After extensive experimentation and model refinement, the researchers found an accuracy of 99% and 96% for disease detection and severity detection, respectively.

As stated in the article [16], this research is aimed at identifying potato leaf disease using deep learning models compared with machine learning. In this study, they collected a total of 5000 images of potato leaves from CFIA, USDA, and potato farms in Ardebil and Iran with five class labels: Black Scurf, Common Scab, Black Leg, Pink Rot, and Healthy. Finally, after extensive experimentation with the models, the CNN model shows exceptional accuracy (99%) compared to learning models such as AlexNet, GoogleNet, VGG16, and R-CNN and machine learning (ML).

This research was aimed an early stage of an orange fruit disease detection using an AI-based technique by comparing CNN, RF, ANN, and SVM [17]. In this work, the researcher collected a total of 1,252 images of orange fruits from the Tigray region, Ethiopia. After training these models, the CNN model achieved the highest performance with an accuracy of 99%, significantly outperforming the other models.

These authors [18] stated that an innovative potato leaf disease detection using an efficient deep learning model. In this article, the researcher trained two class labels of diseases from the existing plant village dataset (early blight and late blight) and the three class labels collected manually, such as potato leaf roll, potato verticillium wilt, and healthy potato. For these datasets, after various experiments and model refinement, the researcher found an accuracy of 97.2% on the testing set.

In this article [19], the researchers investigated the detection and classification of potato leaf diseases using pre-trained transfer learning models by comparing VGG16, VGG19, MobileNetV2, and ResNet50. In this work, they collected the Plant Village dataset sourced from publicly available data and made extensive model refinements, and finally, ResNet50 outperformed the other models with an accuracy of 97%.

Although there have been various researchers who have worked by using pre-trained transfer learning models and attained very high performance accuracy for plant disease detection, even also for potato leaf disease detection, there is a lack of interpretability and explainability of AI (XAI) on the models [8].

Therefore, this research study is proposed to address these gaps by using pre-trained transfer learning models and integrating interpretability and explainability of AI (XAI) on potato leaf disease detection. In this work, we integrated Grad-CAM++ explainable AI (XAI) [11] into our transfer learning model of EfficientNet-B7, which is an enhanced technique of EfficientNet-B0 [10]. This method is not only to evaluate performances but also to foster transparency in real-world agricultural applications.

III. Methodology

A. Dataset And Pre-processing

For brightness and background normalization, we used histogram equalization to enhance the contrast and background subtraction using the technique of Gaussian mixture models (GMM) to separate the potato image from the background, respectively. This research study was done in collaboration with agricultural researchers at Adigrat University, and they provided excellent feedback, which enabled us to continue other work. For this model, we employed a total of 2,152 images of potato leaves, such as early blight (1,000 images), late blight (1,000 images), and healthy (152 images), from the publicly available Plant Village dataset [20]. As stated in Table I, to prevent imbalance in our models, we only increased the dataset size of healthy from 152 images to 1,000 images. Generally, to improve the model's generalization and diversity and prevent overfitting, we considered data augmentation [21]. During this model's training, we employed multiple fundamental data augmentation methods. We used 20% of the degree rotation range, a width and height of 0.2 shift range of magnitude, and a 0.2 magnitude for the shear range and zoom range, and we were considering horizontal and vertical flip. In addition to this, all the images of the potato leaves were resized to 224 x 224 pixels to match image dimensions to models [21] and trained on 10% for testing, 10% for validation, and 80% ([22]) for training, as shown in Table III.

TABLE I
Dataset Sample for Potato Leaves

Disease Class	Before Augmentation	After Augmentation
Healthy	152	1,000
Early blight	1,000	1,000
Late blight	1,000	1,000
Total	2,252	3,000

B. Proposed Model

In our research study, we employed four pre-trained models of EfficientNet-B7, which is a family of EfficientNetB0 [10], VGG19 [12], VGG16 [12], and ResNet50 [13]. These

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models were trained using transfer learning from ImageNet [23], and in this research study, first, we froze these models to leverage the pre-trained weights, and finally, we unfroze these, which enabled us to customize them.Due to this, we customized our number of epochs to 65, we tuned our learning rate and the batch size to 0.0001 and 32, respectively, and we set the dropout of the learning rate to 0.5 to prevent overfitting.

After we customized the models, these models were compared with performance measure classifiers, such as accuracy, recall, precision, and F1 score, and the EfficientNet-B7 model was found to have the highest performance measure of accuracy, recall, precision, and F1 score, as stated in Table IV.

After these models were customized and evaluated, we integrated an explainable AI (XAI) of the Grad-CAM++ technique for EfficientNet-B7 to enhance the interpretation and understanding of the decision-making process of the model as shown in Fig. 10.The comprehensive workflow of the proposed model is shown in Fig. 1.

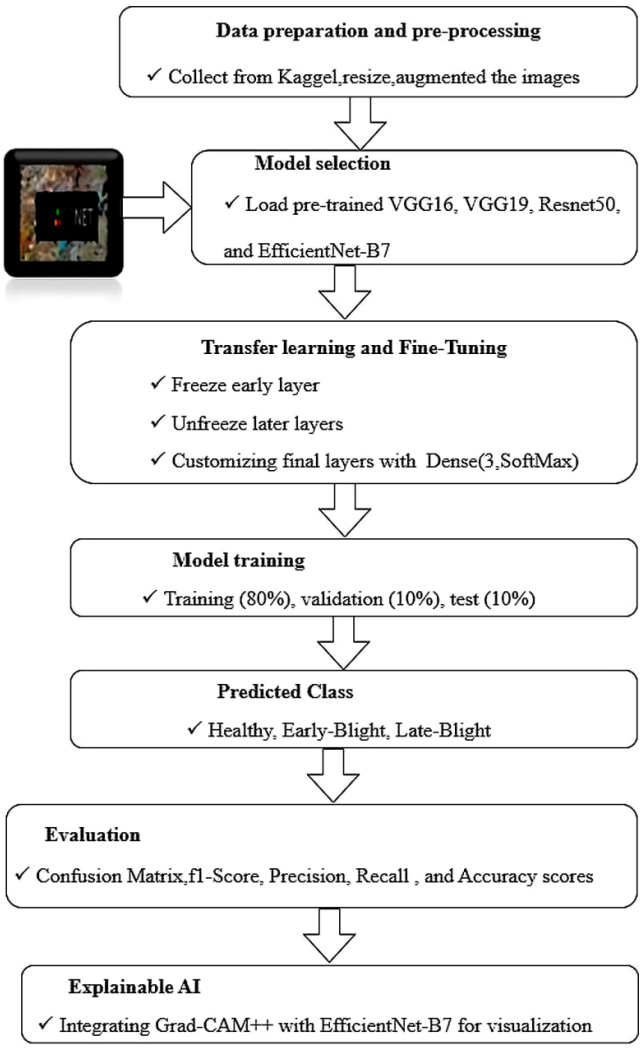

Fig. 1. Proposed Architecture For Potato Leaf Disease Detection.

1) Explainable Artificial intelligent (XAI): This is a field of artificial intelligence that mainly enables stakeholders to understand the reasoning behind AI output and interpret the decision-making process of deep learning models, especially in plant disease detection [24] [8]. Among the well-known types of XAI are Grad-CAM, Grad-CAM++, SHAP, and LIME [11]. In this study, we used Grad-CAM++, which is an advanced visual explanation technique of Grad-CAM [24] [8]. This model provides a visual explanation of the potato leaves, which allows the researchers and professionals to understand which parts of the regions are affected and helps to build trust in the model predictions. To visualize these models, initially, our models are trained on potato leaves, which include 1,000 images of healthy leaves and 1,000 images per class of those affected by early blight and late blight, and then, finally, we saved these models. Once these models are trained and saved, Grad-CAM++ computes the gradients of the predicted class score relative to the feature maps of the last convolutional layer.

C. Environment Requirement

We trained our models with the TensorFlow deep learning library at Google Colab with a T4 GPU-enabled computing system. We employed the T4 GPU computing system in order to greatly reduce the training of our models.

D. Hyper-Parameters Of The Models

Our models (EfficientNet-B7 [10], ResNet50 [13], VGG16 [12], and VGG19 [12]) were trained at the same hyper-parameters and with a total of 3,000 images of potato leaves with loss function as categorical cross-entropy. For these models, to prevent overfitting and underfitting, we tuned the experiment with dropout rates of 0.5. We used the number of epochs to 65 and an early stopping technique to halt training when the validation loss stops improving. For reducing the spatial dimension of these models, we adjusted an experiment with average pooling and employed Softmax as an activation function and the Adam optimizer as a compiler. A comprehensive overview of the hyper-parameters was stated as shown in Table II.

TABLE II
HYPER-PARAMETERS OF THE MODELS

Hyper-parameters	Description
Learning Rate	0.001
Optimizer	Adam
Batch Size	32
Number of Epochs	65
Loss Function	Categorical cross-entropy
Activation Functions	Softmax
Fine-tuning Layers	Some layers frozen, others trainable (specific to architecture)
Dropout Rate	0.5
pooling	Average
Filter and Kernel	65 by 3X3
Early Stopping	Stop if validation loss does not improve for 10 epochs

E. Performance Evaluation And Visualization:

For our models, we use accuracy, recall, precision, and F1 score [25] as performance measures for classifiers as depicted in equations 1-4 and Grad-CAM++ [11] for explainability and visualization purposes.

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN} \tag{1}$$

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$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

where TP,FP,TN,and FN indicating true positive ,false positive,true negative,and false negative,respectively.

IV. Result And Discussion

A. Dataset

This study used a total of 3,000 images of potato leaves, which include early blight (1,000 images), late blight (1,000 images), and healthy (1,000 images), as stated in Table I. In this phase, we split our dataset into 80% for training (2,400 images) and 10% for validation and for testing (300 images), as shown in Table III.

TABLE III
Dataset split for Potato Leaves

Class	Total Images	Training 80%	Validation 10%	Testing 10%
Healthy	1,000	800	100	100
Early Blight	1,000	800	100	100
Late Blight	1,000	800	100	100
Total	**3,000**	**2,400**	**300**	**300**

B. Experimental Results

This section presented detailed results of EfficientNet-B7 [10], ResNet50 [13], VGG16 [12], and VGG19 [12] by considering performance measures of accuracy, recall, precision, and F1 score for potato leaf disease detections, as depicted in Table IV. The highest accuracy is achieved by EfficientNet-B7, as shown in Table IV, indicating its superior ability to distinguish between healthy and diseased potato leaves. It achieved an accuracy of 99.95%. The recall and precision values of healthy were 1.00 and 0.99, respectively, and for early blight, their recall and precision values were 0.99 and 1.00, respectively, and for late blight, their recall and precision values were 1.00 for both. This indicates near-perfect identification of all classes; especially late blight was identified without any error. The F1-scores were 1.00 for the healthy, 0.99 for early blight, and 1.00 for late blight, reflecting high and balanced performance between precision and recall across all classes.

TABLE IV
Experimental ResultS of the models

Model	Class	Matrics		
		Precision	Recall	F1 Score
EfficientNet-B7	Healthy	0.99	1.00	1.00
	Early blight	1.00	0.99	0.99
	Late blight	1.00	1.00	1.00
Test Accuracy (EfficientNet-B7)		99.95%		
VGG19	Healthy	0.97	0.99	0.98
	Early blight	1.00	0.99	0.99
	Late blight	0.98	0.97	0.97
Test Accuracy (VGG19)		98%		
VGG16	Healthy	0.99	0.97	0.97
	Early blight	1.00	0.96	0.96
	Late blight	0.93	0.99	0.97
Test Accuracy (VGG16)		97.1%		
ResNet50	Healthy	0.99	0.96	0.97
	Early blight	1.00	0.96	0.96
	Late blight	0.93	0.99	0.96
Test Accuracy (ResNet50)		97%		

VGG19 followed EfficientNet-B7, demonstrating excellent generalization, with an accuracy of approximately 98%, as shown in Table IV.

VGG16 also performed less accurately than VGG19 (97.1%), as depicted in Table IV. As shown in Table IV, ResNet50 showed the lowest performance with an accuracy of 97%, almost the same as VGG16.

C. Confusion Matrix Analysis

To improve these models, we used the Adam optimizer as a compiler, we included an early stop to terminate the training process if there was no improvement with a difference of 10 epochs, and as a classifier, categorical cross-entropy was considered.Fig. 2, 3, 4, and 5 show the confusion metrics of our models. As stated in Fig. 2, the highest confusion matrix was found in EfficientNet-B7 [10].

D. Training and Validation Performance

Fig. 6, 7, 8, and 9 show the training and validation loss and accuracy of our models on the number of epochs 65. Fig. 6 achieved a training accuracy of approximately 99.95%, indicating excellent learning of training in potato leaf disease detection, and the training loss decreased steadily, reaching 0.0094. Its validation accuracy is 100%, demonstrating the model's strong generalization capability for unseen potato disease detection, and the validation loss stabilized around 0.0065, suggesting minimal overfitting.

E. Explainable Artificial Intelligence(XAI)

After we compared our models, we selected EfficientNet-B7 due to its superiority in accuracy, recall, precision, F1 score, and efficiency in potato leaf disease detection, as shown in Table IV. To enhance visualization, understanding, and interpretation, we integrated an explainable artificial intelligence (XAI) of the Grad-CAM++ technique [11] for EfficientNet-B7 [10], which provides better localization of disease regions by capturing multiple important areas in the image of the potato leaf as depicted by fig. 10.

F. Comparative Analysis with Previous Work

In this section, we compared the performance of EfficientNet-B7 with its explainable AI (XAI) of the Grad-CAM++ technique with a previous research study of potato disease detection. Therefore, compared to earlier works, EfficientNet-B7 with the Grad-CAM++ technique achieved the highest performance, with a test accuracy of 99.95%, and earlier works were depending solely on accuracy matrices such as accuracy, recall, precision, and F1 score and not considering explainable AI. See the details in Table V and Fig. 10.

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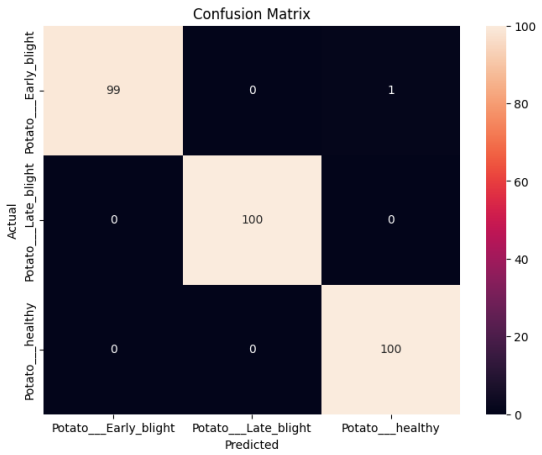

Fig. 2. EfficientNet-B7

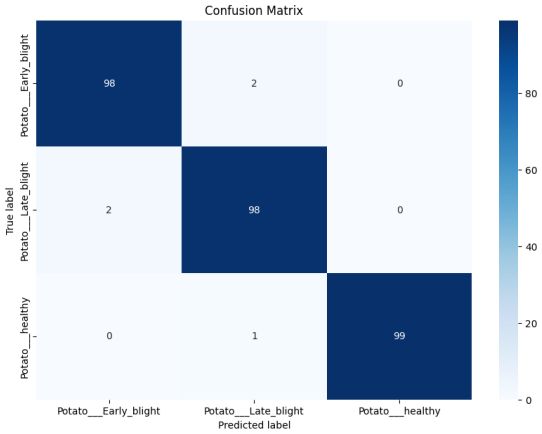

Fig. 3. VGG19

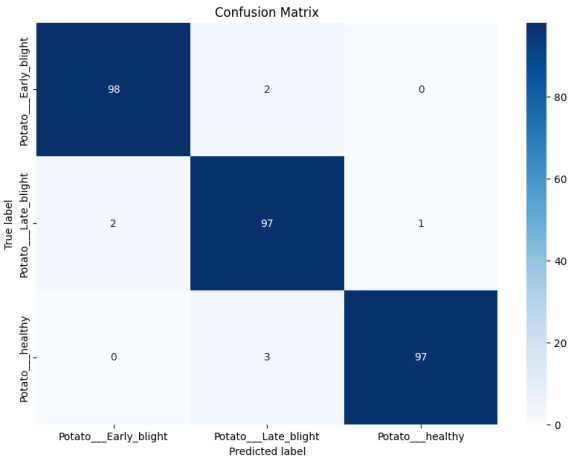

Fig. 4. VGG16

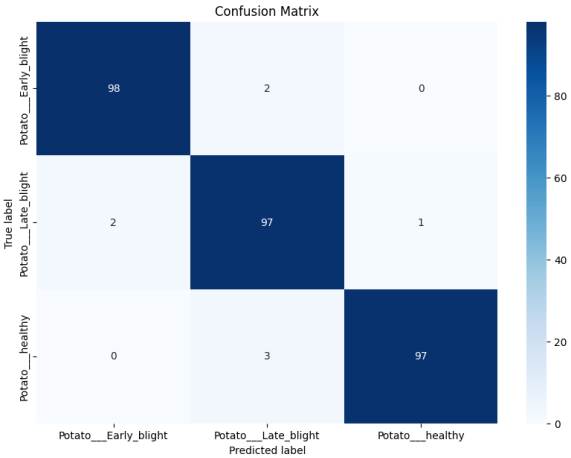

Fig. 5. ResNet50

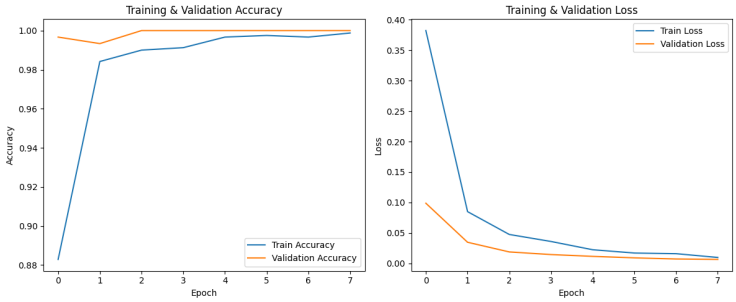

Fig. 6. EfficientNet-B7

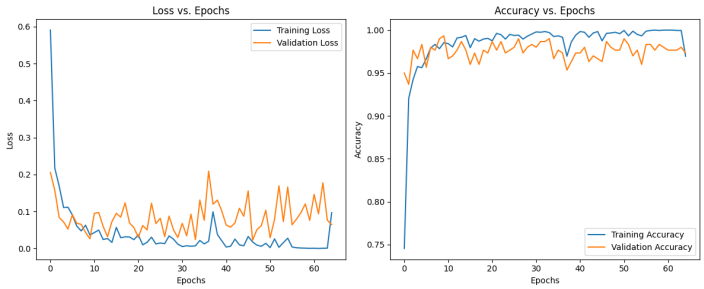

Fig. 7. VGG19

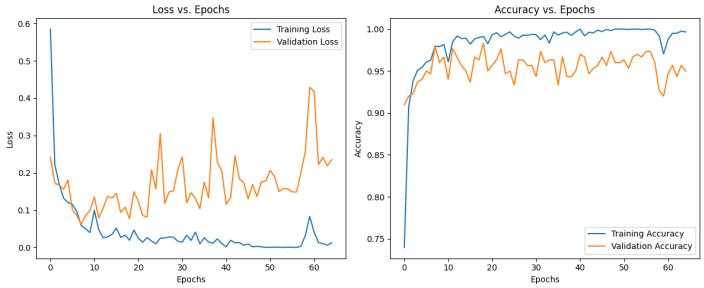

Fig. 8. VGG16

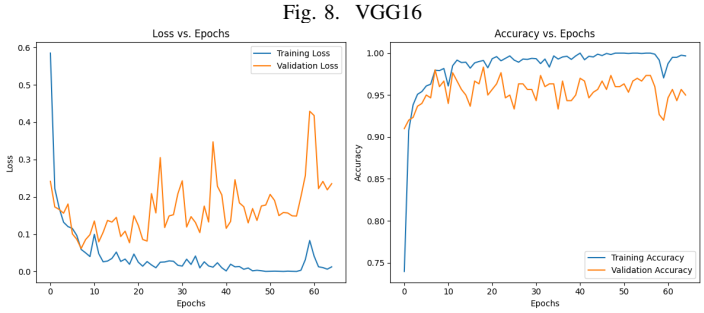

Fig. 9. ResNet50

TABLE V
COMPARATIVE ANALYSIS

Authors/Year	**Accuracy (%)**	**Methods**	**XAI**
[16] (2022)	99.0%	CNN	No
[18] (2023)	97.2%	EfficientNet	No
[17] (2024)	99.0%	CNN	No
[15] (2025)	99.0%	AlexNet, VGG16	No
[14] (2025)	98.0%	VGG16	**Grad-CAM**
[19] (2024)	94.0, 90.0, 95.0, 97.0	VGG16, VGG19, MobileNetV2, ResNet50	No
Proposed (2025)	**99.95%**	**EfficientNet-B7**	**Grad-CAM++**

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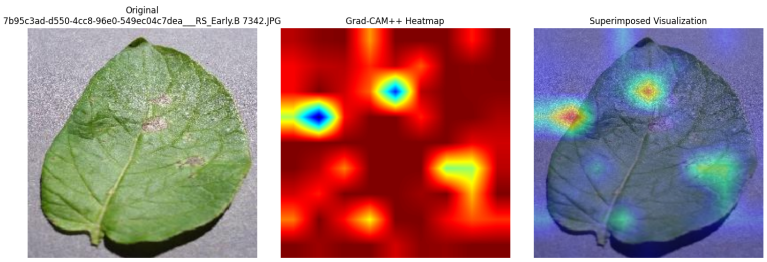

Fig. 10. Potato Disease Detection Using XAI of Grad-CAM++

V. Conclusion And Future Recommendation

Our research study focused on potato leaf disease detection using pre-trained transfer learning models of EfficientNet-B7, ResNet50, VGG16, and VGG19, which were implemented on Google Colab with a T4 GPU. In this work, we trained our models with the same hyper-parameters and with a total dataset of 3,000 images (1,000 early blight, 1,000 late blight, and 1,000 healthy) sourced from public availability. After extensive training of these models, we found test accuracy of 99.95% for EfficientNet-B7, 98% for VGG19, 97.1% for VGG16, and 97% for ResNet50. Therefore, we selected EfficientNet-B7 due to its superiority in accuracy, recall, precision, and F1 scores among the other models. Finally, to enhance the interpretability and explainability of the EfficientNet-B7 model, we integrated explainable AI (XAI) of the Grad-CAM++ visualization technique. This method is not only to enhance its performance accuracy but also fosters robustness, truth, and decision-making processes. Since our models were developed on Google Colab with GPU, which has high performance for processing, to be easily accessed by the farmers, it is recommended to integrate these models with either a mobile application or a web-based application. To enhance visualization, understanding, and interpretability, we recommended to future researchers to incorporate other explainability techniques such as Grad-CAM, SHAP, LIME, and saliency maps [11] and recommended to include IOT and remote sensing technology for comprehensive potato leaf disease detection.

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